



Verify

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EasyForensicspicoCTF 2024grepbrowser_webshell_solvablechecksum

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Description

People keep trying to trick my players with imitation flags. I want to make sure they get the real thing! I'm going to provide the SHA-256 hash and a decrypt script to help you know that my flags are legitimate.

Additional details will be available after launching your challenge instance.

This challenge launches an instance on demand.

Its current status is: **NOT_RUNNING**

Launch Instance

Hints ?

123

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 picoCTF{FLAG}

Submit Flag

Author: The Analyst: Hyposelenia

Challenge: Verify

Category: Forensics

Date: 04/02/26



I. Objective

The objective of this challenge is to understand how to verify file integrity using **hash functions**.

The goal is to identify the correct file by matching its hash value and use it to **retrieve the flag**.

II. Background

In cybersecurity, files are often verified using something called a **hash**.

A hash is:

A unique “fingerprint” of a file

Even if a file changes slightly, its hash will completely change.

In this challenge:

- A correct hash was given
- Many files were present
- Only ONE file matched the correct hash
- That file was used to retrieve the flag

III. Tool Used

- **Kali Linux (Oracle VirtualBox)** - Used as the working environment
- **Linux Terminal** - Used to execute commands
- **SSH** - Used to connect to the remote challenge instance

IV. Methodology

1. Launched **Kali Linux** via Oracle VirtualBox
2. Started the challenge instance



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Launch Instance

3. Connected using SSH:

`ssh -p <port> <user>@<host>`

```
root@kali:~/Downloads
$ ssh -p 57924 ctf-player@rhea.picocftf.net
The authenticity of host '[rhea.picocftf.net]:57924 ([3.136.191.228]:57924)' can't be established
ED25519 key fingerprint is SHA256:QKdv+RGJL0UYRDM66IiGQ5dsXOX7DQFqB7umTylh+IU.
This key is not known by any other names.
```

4. Typed yes

```
ED25519 key fingerprint is SHA256:QKdv+RGJL0UYRDM66IiGQ5dsXOX7DQFqB7umTylh+IU.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '[rhea.picocftf.net]:57924' (ED25519) to the list of known hosts.
```

5. Entered the provided password (*input is hidden*)

```
ctf-player@rhea.picocftf.net's password:
Welcome to Ubuntu 20.04.3 LTS (GNU/Linux 6.8.0-1021-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/advantage

This system has been minimized by removing packages and content that are
not required on a system that users do not log into.

To restore this content, you can run the 'unminimize' command.

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.
```

6. Listed files via `ls` command

```
ctf-player@pico-chall$ ls
checksum.txt  decrypt.sh  files
```



7. Read the checksum file:

cat checksum.txt

```
checksum.txt decrypt.sh files
ctf-player@pico-chall$ cat checksum.txt
3ad37ed6c5ab81d31e4c94ae611e0adf2e9e3e6bee55804ebc7f386283e366a4
```

8. Observed a **SHA256** hash value

9. Checked files inside directory

ls files/

```
ctf-player@pico-chall$ ls files/
8agq1PLS 3Br1DAb0 6vY68JA ABvRy5U DH010q0q G8MIUPTw J75ycvvy LRXQ50hw Qg8KRab0 Q8P8w8ua TSJCKwIT V2C27uy0 XFZqFKg1 ZFRDLvRe b1ZJH0Ww dkV8p1DF g0QD13TI k1FFh0c1 gR2nzVaf PU0Hr5vR vFW94Ek3 yZqsPnuQ
8p8Kv2d5 3PmKdHHH 7U4d5T0L Agq56rWn DM051bHe G0dH0Qto J8V7XHTa LQVUw70 02UAGP3p Q13CHaNg TTJL07LQ VHV1U0e8f XJ2vKSLg ZgC80WdC C30E4AtP d1C00206 gY52Y8T0 1a1P16dy 0271CyG0 rFNg00L vnr7VUto 2M4q3Lw2
8w8A1ot 3eK0Zcy 7cZ209p BMLG0im DM0U1Fah G0dH0Qto J30080C MVL0E0C P02523H R1J0MP20 The0n0cy Y0P0uq1b XrAsL2L1 20S8a47 00a0a0w 00100574 00101VYV 1U00Q240 0a0P2000 r0W5G0U w0333Jm 2M5A1L3
8vYz22am 3eB8VesU 63Myz1LP 8P1e0u00 D17h281J H3rL0pMl 3K7008Rl Mge8V6X7 P83Mh3nc R2J2lp0X TqHf2H54 Vj67lQyx XvYFY0dG aN010e5X ca5exAe8 e710V0B1 h0YxNGC2 02K011H0 0e08Cw04 r0dFR0Mn wV0aJCep 2M01n0ew
120UEF40 3K5ZW04N 80w7QTAA BM0Rgk48 D13K48PL H010K0m 31150Y11 M0fNVxMB P8h0NfB8 RXC0K0uy Tui6wJfr W85tW8P8 Kw38PYK0 0wRn0GvY cdL3dv1c exG0tYty h0D0eCw0 m5C04h5 0rc2TmK0 s1055Khc u333M0u 2W6R0AcB
12R70d0h 3KVAJ1TX 0w0e10G0 20h1L395 D0LY0d0h H0P0R1002 K101C0P2 A0d0F20w P08R0Y3p R2f0eW0w T0w0p0F2 R0T0e0r01 X0u0e0r0 0K0e0000 ch0d0k0p 000040E1 1R0q005L mV0E220W 0271D0Z0 0010M0R0 w7505P0 220F010h
1E0NR0V1 30K1111v 80S0tV0U Bv0LEv2R E5U070JC H05T1100 K000Q08B NFe06PFS PK1Cf0R0 Ricc0X0h U100Y020 W000B205 Y0U0W0E0 0470xT0e c1Y0130p F30Y210k 10k0Z0W0 0cJ0gR0R pyJ5Kx20 13YK0d1U w1P0W0Z0 2H01E08C
1FJaH53f 3XJxU50P 80q08FV5 C560J0D1 EAZM100M H0FFZ0CW KCL7E0H0 NgY5Gyng PUL0V0R0x RuvyH0VU 00VYQ3H0 Wz1AF0P0 YCF9V0P0R D703V1Z8 00e0J0bu f7028T0e 11R0FF0K 005211W0 11209M0L 1L09H0M0 0t000V77
1cV0Y0BL 49gLn120 920k3PFX C0J1F1PS E086057K H1W0K50D KVL0F22V N0p1Z0m0 P321S0AC 59J3AP0R 0K1454V0 WxFlV0B0 VG1pC0D0 001bV0M8 00w0B0ER FwZ0U0CF 35T1C1Z62 n7C1B0PK 02Y7F0U0 02J05W06 W03RFP7B
131Q00R 49P0R01 920R000P C0J140V 0V0L1V0V H0U02C0C K000P20K Q100M13H S030V21Z 0M0000Jb X051L0G 0M0000Jb Y0H100G0 0111500K C043C02U 0R0040W 3M00W0M 00J03100 01000002 02001000 02001000
200Q0H2C 5p130chp 901U085K C030405K E10U0M0R 17V7J05A K0M0V0D0 0V2K0R0X P0Y2K0R5 S01V0N2T U01C10BF X0010W0Y V0d10H0V Df00c1Z1 d1N000V FFM030V 35NC00U0 0F0E10A6 00051K1h V0L0M0L1 Y0E0W0K0
200M0CT 50605Tg A00K000y C0tN003K F0G1L0PC 1A1KR0Wf 140K110c 0K3110B9 P2U1J0R0 50F1V05J 010P1013 X0C0U1M0 Y0d0J0T10 b110g0R7 0520010U FFM00W0Y 02001000 00000000 00000000 00000000 00000000 00000000 00000000
220010Q 00P0Y10D 40F1J0H0 C00Y021V F0W0J227 1H00L0L0 L00C1105 0P0W00F0 Q11000V0 30Q21002 UJ0F2002 X0T00003 Z0K0J00g 00000030 0V07100T fJ00T020 03000000 00000000 00000000 00000000 00000000 00000000
300N00W5 0R00V10K A00000Vt C212F00C 00M1000H 1000h0V V1000010 0P0310P0 0000B0L0 T00Y1000 00F100F0 X000D05b 20N00L00 00000000 02V00000 01N000F0 K00100J0 000000VY 00P0F010 V10T01L0 Y105L001
```

10. Generated hashes and searched for match:

sha256sum files/ | grep <hash_value>*

```
ctf-player@pico-chall$ cat checksum.txt
cat: checksum.txt: No such file or directory
ctf-player@pico-chall$ cat checksum.txt
3ad37ed6c5ab81d31e4c94ae611e0adf2e9e3e6bee55804ebc7f386283e366a4
ctf-player@pico-chall$ sha256sum files/* | grep 3ad37ed6c5ab81d31e4c94ae611e0adf2e9e3e6bee55804ebc7f386283e366a4
3ad37ed6c5ab81d31e4c94ae611e0adf2e9e3e6bee55804ebc7f386283e366a4 files/e018b574
```

11. Identified the correct file

```
cat: checksum.txt: No such file or directory
ctf-player@pico-chall$ cat checksum.txt
3ad37ed6c5ab81d31e4c94ae611e0adf2e9e3e6bee55804ebc7f386283e366a4
ctf-player@pico-chall$ sha256sum files/* | grep 3ad37ed6c5ab81d31e4c94ae611e0adf2e9e3e6bee55804ebc7f386283e366a4
3ad37ed6c5ab81d31e4c94ae611e0adf2e9e3e6bee55804ebc7f386283e366a4 files/e018b574
```

12. Ran the decrypt script:

./decrypt.sh files/<correct_file>

```
ctf-player@pico-chall$ ./decrypt.sh files/e018b574
picoCTF{trust_but_verify_e018b574}
ctf-player@pico-chall$ ls files/
```

13. Flag is retrieved



V. Result

picoCTF{trust_but_verify_e018b574}

```
ctf-player@pico-chall$ ./decrypt.sh files/e018b574
picoCTF{trust_but_verify_e018b574}
```

VI. Explanation

What is a Hash?

A hash is:

A unique code that represents a file

Example:

File → turns into:

3ad37ed6c5ab81d31e4c94ae611e0adf...

Think of it like:

- Your fingerprint
- No two people have the same one

What is SHA256?

SHA256 is:

A type of hash function

It takes any file and turns it into a long code

Important:

- Same file → same hash
- Different file → different hash

Why do we use hashes?

To:

- verify files
- check integrity



- detect tampering

Real-life analogy:

You send a file

You also send its hash

Receiver checks:

- If hash matches → file is safe
- If not → file was changed

What does this command do?

`sha256sum files/*`

It:

- checks all files
- generates hash for each

What does grep do?

`grep <hash>`

It means:

“Find this exact text”

Combined command:

`sha256sum files/* | grep <hash>`

Means:

“Find which file has THIS hash”

What is ./decrypt.sh?

This is a script that:

Uses the correct file to unlock the flag

Think of it like:



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- Many keys
- Only ONE opens the lock

VII. Conclusion

This challenge demonstrates the importance of verifying data using hash functions.

By comparing file hashes and identifying the correct match, the appropriate file was used to retrieve the flag.

This reinforces a key concept in cybersecurity:

Always verify data integrity — trust, but verify.

— The Analyst: Hyposelenia